SEMINARREPORT

A seminar Report on

Smart Bill Book

Submitted in partial fulfillment for the award of

THIRD YEAR DIPLOMA IN

COMPUTER ENGINEERING

By

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FOR THE ACADEMIC YEAR : 2026 – 2027

DEPARTMENT OF COMPUTER ENGINEERING S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

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(An Affiliated Institute of Maharashtra State Board of Technical Education)

DEPARTMENT OF COMPUTER ENGINEERING

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ABSTRACT

The increasing use of digital technologies has made it important for businesses to manage their daily operations efficiently. Traditional billing methods used in gas agencies involve manual calculations, paperwork, and handwritten records, which can lead to errors, time consumption, and difficulties in maintaining customer information. The Smart Bill Book for Gas Agency is a computerized billing system designed to simplify and improve the billing process.

This project report presents a review of twelve research papers related to billing systems, gas agency management, automated invoice generation, customer record management, and computerized billing applications. The reviewed studies focus on billing accuracy, customer information management, automated bill generation, and digital record maintenance. The report discusses the working principles, system architecture, advantages, and limitations of existing billing systems.

The report concludes with a project initiation plan for developing a Smart Bill Book for Gas Agency that can simplify daily billing activities, reduce paperwork, and support the effective management of customer and billing records.

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CHAPTER 1 INTRODUCTION

Billing management is an important part of every business, including gas agencies. Gas agencies handle daily activities such as managing customer information, maintaining cylinder delivery records, preparing bills, and tracking billing transactions. Manual billing methods can be timeconsuming and may lead to calculation errors, duplicate entries, and difficulties in maintaining records.

Traditional billing methods often depend on handwritten registers and manual calculations, making it difficult to retrieve previous billing information and update customer details. As the number of customers increases, managing these activities becomes more complicated. Therefore, a computerized billing system is needed to simplify operations and improve efficiency.

The Smart Bill Book for Gas Agency is a computerized billing system designed to make billing easier, faster, and more organized. It provides a digital platform for managing customer details, generating bills, calculating billing amounts, and maintaining billing records. The system helps reduce paperwork, minimize manual errors, and save time.

The main objective of this project is to provide a simple and user-friendly solution for billing and record management. It helps gas agency staff store, retrieve, and manage customer information efficiently, improving the accuracy and organization of daily billing operations.

This report reviews twelve research papers related to gas agency management, computerized billing systems, automated invoice generation, and digital record management. It also discusses existing systems, their limitations, the proposed system, its objectives, scope, advantages, and development plan.

Overall, the Smart Bill Book for Gas Agency aims to improve traditional billing practices by providing an efficient and organized approach to managing daily billing activities.

CHAPTER 2 LITERATURE REVIEW

SMART BILL BOOK – GAS AGENCY MANAGEMENT SYSTEM

1. Customer Registration and Authentication

Smart Bill Book provides a customer registration and login system that allows users to create accounts using their personal information. Customers can register by entering their name, mobile number, address, email, date of birth, and LPG connection type. The system stores customer details in a database and uses login credentials to authenticate registered users. This feature helps maintain organized customer records and provides controlled access to the website.

2. LPG Connection Management

The system allows customers to apply for new LPG connections through an online platform. Customers can select domestic or commercial connections according to their requirements. They can submit the necessary information and documents for verification. Agency employees can review connection requests and update their application status. This feature reduces paperwork and simplifies the LPG connection process.

3. Online Gas Cylinder Booking

Smart Bill Book provides an online LPG cylinder booking facility for registered customers. Customers can request cylinder refills without physically visiting the gas agency. The system maintains booking information, including customer details, booking dates, and current booking status. Agency employees can view and process booking requests through their dashboard. This feature improves convenience and reduces manual booking operations.

4. Admin and Employee Management

The website provides separate login systems and dashboards for administrators and employees. Administrators can manage employee accounts, customer information, and gas agency operations. Employees can access assigned activities such as cylinder bookings, customer requests, and delivery updates. Role-based access helps organize responsibilities and restrict access to administrative functions. This improves coordination between agency staff members.

5. Automated Billing and Invoice Generation

Smart Bill Book includes automated billing functionality to simplify financial transactions within the gas agency. The system can generate bills and tax invoices based on cylinder bookings and applicable prices. Customers can access billing information and download available invoices. The billing records are maintained in the database for future reference. This feature reduces manual calculations and improves billing accuracy.

6. LPG Demand Prediction Using Machine Learning

This additional feature is designed to predict future LPG cylinder demand using historical booking records. Machine Learning algorithms can analyze previous cylinder bookings, seasonal patterns, and customer demand trends. The predicted results can help gas agencies estimate future cylinder requirements. This supports inventory planning and reduces the possibility of stock shortages.
Demand forecasting can also improve overall agency efficiency.

7. Smart Delivery Management

The smart delivery management feature helps gas agencies organize and monitor LPG cylinder deliveries. Employees can update delivery statuses and maintain records of completed and pending deliveries. The system can analyze delivery performance to identify delays and operational problems. These insights can help improve delivery scheduling and resource allocation. This feature aims to provide faster and more reliable cylinder delivery services.

8. LPG Stock and Inventory Management

Smart Bill Book can help gas agencies maintain records of available LPG cylinders and inventory movements. The system can track cylinder stock, completed deliveries, and pending booking requirements. Inventory information allows administrators to monitor cylinder availability. Lowstock alerts can be introduced to notify managers when additional cylinders are required. This feature helps reduce inventory shortages and supports efficient stock management.

9. Customer Feedback and Complaint Management

The system can provide a facility for customers to submit feedback and complaints regarding LPG services. Customers may report delivery delays, booking problems, billing errors, or other servicerelated issues. Employees can review complaints and update their resolution status.

Administrators can monitor complaint records to identify frequently reported problems. This feature helps improve customer satisfaction and service quality.

10. AI-Based Decision Support Dashboard

The proposed AI-based dashboard can assist gas agency administrators in making better operational decisions. It can display information related to LPG demand predictions, cylinder bookings, delivery performance, and inventory levels. Machine Learning results can be used to identify business trends and potential operational issues. The dashboard can also provide alerts and recommendations based on available data. This feature aims to improve planning and overall agency management.

11. Automated Reports and Business Analytic

Smart Bill Book can generate business reports related to customer registrations, cylinder bookings, billing records, and delivery activities. These reports help administrators understand the daily and monthly performance of the gas agency. Data analysis techniques can identify booking trends, revenue patterns, and customer service requirements. Reports can be presented through tables and graphical representations. This feature supports business monitoring and administrative decisionmaking.

12. LPG Booking Status Tracking

The booking status tracking feature allows customers to monitor the progress of their cylinder booking requests. Booking records can display different stages, such as pending, approved, dispatched, and delivered. Agency employees can update booking statuses as cylinder requests are processed. Customers can view the latest available information through their dashboard. This feature improves transparency and reduces the need for repeated customer inquiries.

13. Digital KYC and Document Verification

Smart Bill Book provides a digital approach to collecting and managing customer KYC information. Customers can enter personal details and upload required documents while applying for LPG connections. Authorized agency employees can review the submitted information and verify customer records. The system can maintain document submission and verification statuses.
This feature reduces paperwork and helps organize customer documentation.

14. Customer Dashboard and Self-Service Portal

The customer dashboard provides a centralized interface for accessing LPG-related services. Customers can view connection requests, booking information, billing records, and available documents. The dashboard can provide options for downloading subscription vouchers and tax invoices. Customers can also monitor their application and booking statuses. This feature improves accessibility and allows customers to manage their LPG services conveniently.

15. Data Security and Database Management

Smart Bill Book uses SQLite for storing and managing information related to customers, employees, LPG connections, bookings, and billing records. Python handles database operations such as inserting, updating, retrieving, and deleting records. Role-based authentication helps restrict access to sensitive administrative functions. Appropriate password protection and input validation can further improve system security. This feature supports organized data storage and reliable information management.

Table No 2.1: Literature review:

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	Computerized Gas Agency Management System	Database Management System, application software	Customer details, stock records and billing information	To replace manual billing and stock maintenance with a computerized system.	Requires accurate data entry and regular database updates.
2	Automated Gas Agency Billing System	Database, billing software	Customer records, cylinder prices and billing transactions	To simplify bill preparation and reduce manual calculation errors	Incorrect customer or price information can result in billing errors
3	Gas Booking and Customer Management System	Web application, database	Customer details, booking records and delivery dates	To manage customer information and gas booking records efficiently	Booking and delivery information must be updated regularly
4	Online Gas Booking and Payment Integration	Web technologies, database, payment gateway	Booking details, payment records and stock information	To integrate online booking, payment processing and stock management	Depends on internet availability and secure payment integration
5	Inventory and Stock Management System	Database Management System, inventory software	Cylinder stock, available quantities and stock transactions	To maintain accurate stock records and monitor cylinder availability	Incorrect stock entries may lead to discrepancies
6	Customer Databse Management System	Relational database, SQL	Customer IDs, names, addresses and contact information	To store customer information systematically and retrieve records quickly	Requires access control, data validation and database backups

7	Digital Invoice Generation System	Billing application, database	Invoice numbers, customer details, prices and totals	To generate digital invoices and maintain billing history	Incorrect pricing rules may affect invoice accuracy
8	Transaction Managemant System	Database, SQL, transaction processing	Payment details, transaction dates and bill records	To maintain transaction history abd support efficient billing record retrieval	Duplicate or incomplete records may affect financial reporting
9	Web-Based Gas Agency Management System	HTML, CSS, JavaScript, server-side programming and database	Customer, booking inventory and billing records	To provide a centralized system for managing gas agency activities	Requires software maintenance, security measures and reliable hosting
10	Automated Bill Calculation System	Programming language, database and calculation logic	Cylinder price, quantity and applicable charges	To calculate bill amounts automatically and reduce repetitive manual calculations	The system must handle incorrect inputs and calculation exceptions
11	Billing Report and Record Management System	Database, reporting tools and application software	Daily bills, monthly transactions and customer records	To generate billing reports and simplify the analysis of transaction records	Report accuracy depemds on complete and consistent data
12	Smart Bill Book for Gas Agency	Web technologies, database and automated billing logic	Customer details, billing records, payment information and cylinder booking data	To develop a digital billing system that simplifies bill generation, customer record management and transaction tracking for a gas agency	Requires proper testing, secure data storage and regular maintenance

CHAPTER 3

1. Overview of Smart Bill Book:

Smart Bill Book is a web-based Gas Agency Management System developed using Python, Streamlit, and SQLite. It is designed to simplify and digitalize the daily operations of LPG gas agencies. The system provides facilities such as customer registration, LPG connection management, cylinder booking, automated billing, and delivery tracking. It also provides separate dashboards for administrators, employees, and customers, making gas agency operations more organized, efficient, and convenient.

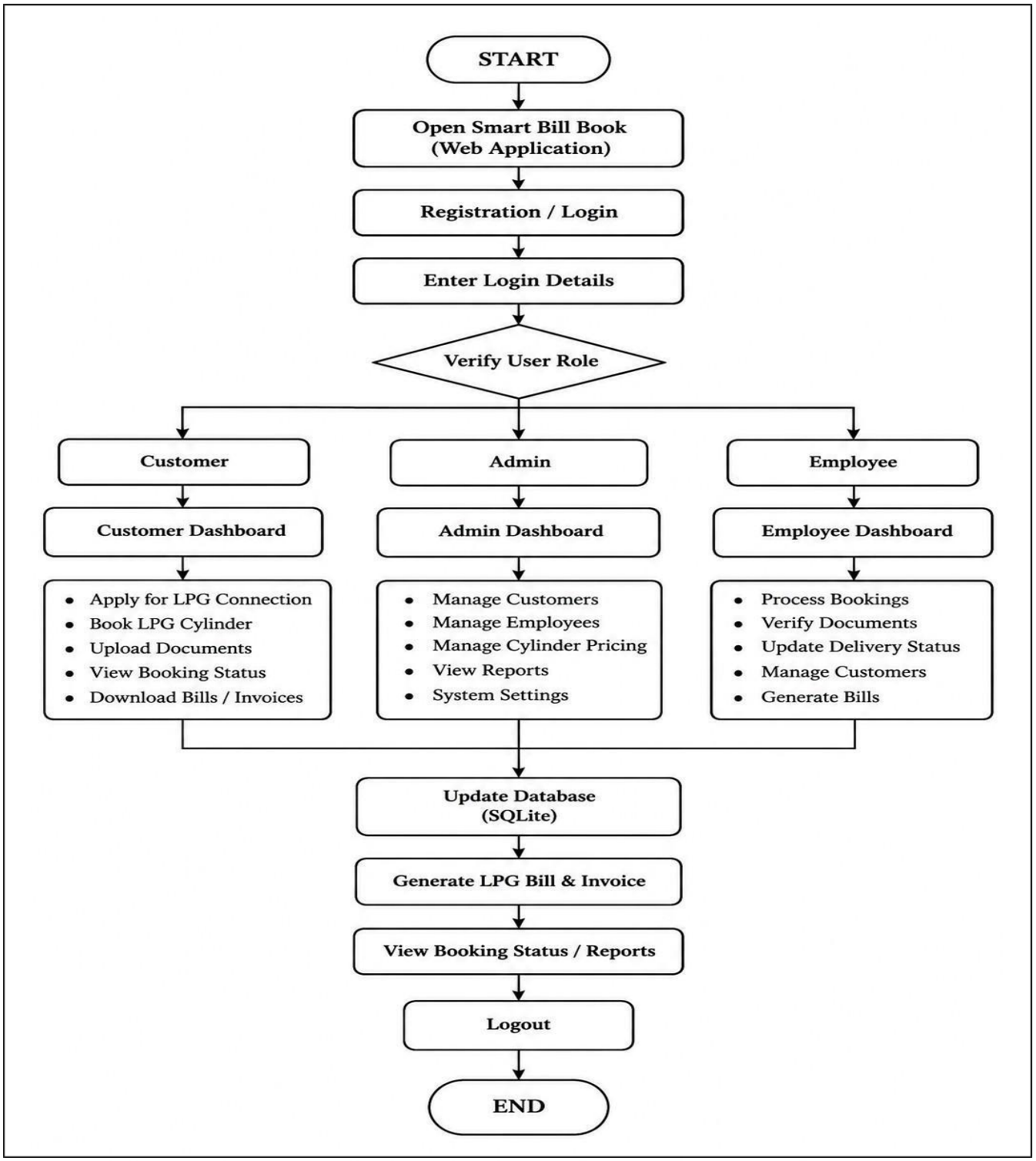
2. Need for Smart Bill Book:

Traditional gas agency management involves manual paperwork, customer record maintenance, cylinder booking registers, and billing calculations. These activities can be time-consuming and may lead to errors, delays, and difficulties in maintaining accurate records. Smart Bill Book aims to overcome these problems by providing a centralized digital platform for managing gas agency operations. The system reduces manual work, improves data accuracy, saves time, and provides customers with easier access to LPG-related services.

3. Objectives of Smart Bill Book:

The main objective of Smart Bill Book is to develop an efficient and user-friendly web application for managing LPG gas agency activities. The system aims to automate customer registration, LPG connection requests, cylinder bookings, billing, and delivery management. It also supports organized database management and role-based access for administrators, employees, and customers. Additionally, the project proposes the use of Artificial Intelligence and Machine Learning for LPG demand prediction, business analysis, and improved decision-making. These features aim to increase operational efficiency, reduce paperwork, and improve customer satisfaction.

4.Working Principle:



3.1 Working of Smart Bill Book.

The Working of Smart Bill Book consists of the following steps:

1. **Start:** The process begins when a user opens the Smart Bill Book Gas Agency Management System through a web browser.

2. **Open Smart Bill Book:** The system displays the homepage containing login and registration options for customers, administrators, and employees.

3. **Customer Registration:** New customers register by entering their name, date of birth, mobile number, address, email, password, and LPG connection type.

4. **User Authentication:** Registered users enter their login credentials, which are verified using the information stored in the database.

5. **Identify User Role:** The system identifies whether the logged-in user is a customer, administrator, or employee and provides access to the appropriate dashboard.

6. **Customer Dashboard:** Customers can access LPG connection requests, cylinder bookings, document uploads, booking information, and available billing records.

7. **LPG Connection Request:** Customers can apply for domestic or commercial LPG connections by submitting the required personal information and documents.

8. **Document Verification:** Authorized agency staff review the submitted customer documents and update the LPG connection application status.

9. **Cylinder Booking:** Customers can submit LPG cylinder booking requests, which are recorded in the database for further processing.

10. **Admin Dashboard:** The administrator manages customer information, employee accounts, cylinder pricing, agency records, and other administrative activities.

11. **Employee Dashboard:** Employees manage LPG booking requests, review customer details, update booking statuses, and coordinate delivery-related activities.

12. **Delivery Management:** Authorized staff can manage cylinder deliveries and update their status to maintain accurate delivery records.

13. **Automated Billing:** The system generates LPG bills and invoices using booking details and applicable cylinder prices.

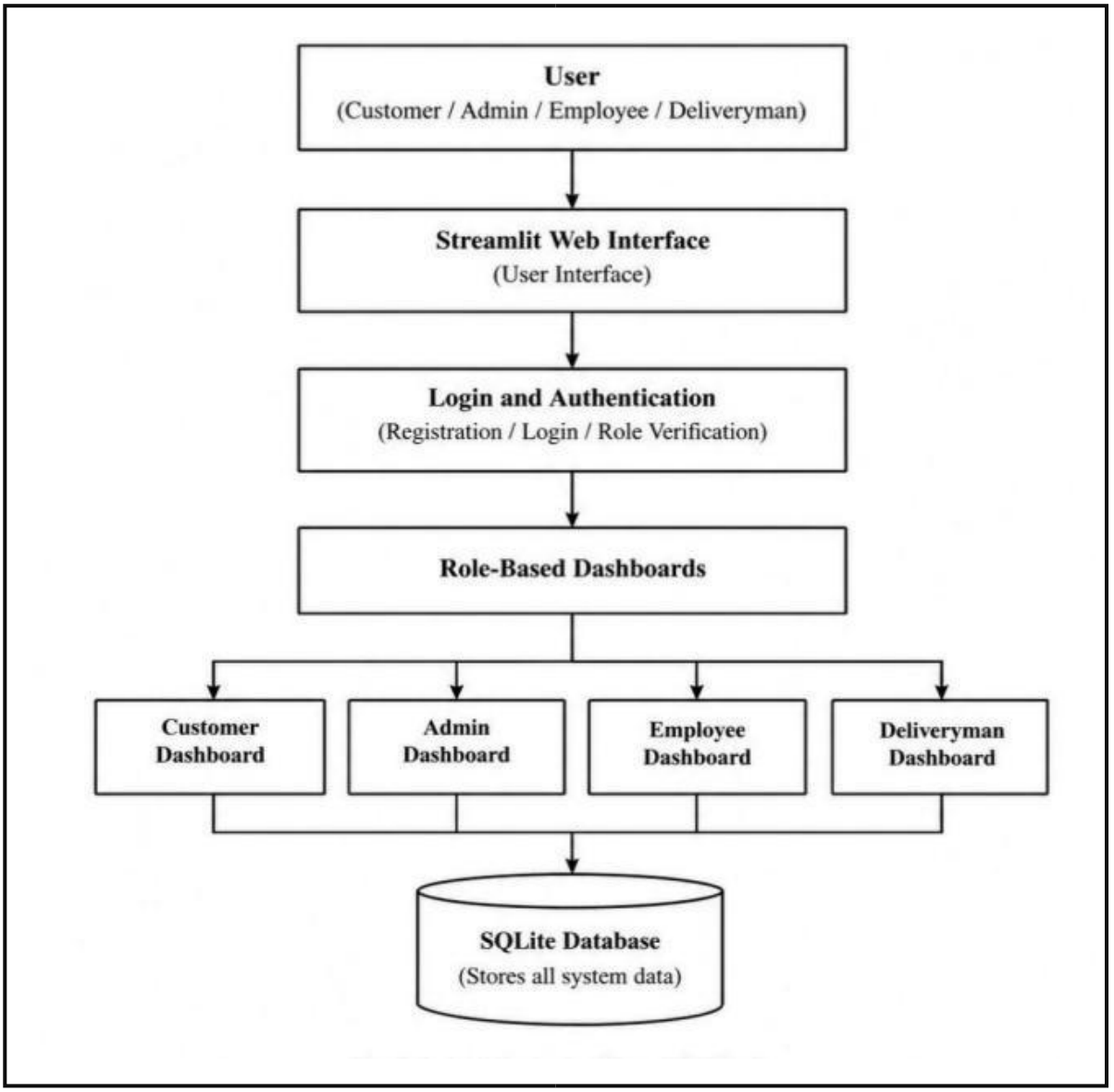
14. Database Management: Customer registrations, LPG connections, bookings, billing information, and other operational records are maintained using SQLite.

15. Reports and Records: Administrators can access available sales, billing, and operational records to monitor gas agency activities.

16. Logout: After completing their activities, users can log out of the system, ending their authenticated session.

17. End: The working process ends after the user successfully logs out of the Smart Bill Book website.\

5.System Architecture and Main Components:



3.2 Architecture of Smart Bill Book.

The Components Are:

The main components of Smart Bill Book are the User Interface, Login and Authentication Module, Customer Dashboard, Admin Dashboard, Employee Dashboard, Deliveryman Dashboard, LPG Connection Management, Cylinder Booking Management, Automated Billing Module, and SQLite Database. The user interface allows customers and staff to access the website, while the authentication module verifies login credentials and user roles. The management modules handle customer records, LPG bookings, document verification, cylinder deliveries, billing, and reports. All these components work together to provide an efficient and organized Gas Agency Management System.

5. Technologies and Techniques Used:

Smart Bill Book is developed using Python for backend programming, Streamlit for designing the web-based interface, and SQLite for storing customer, employee, booking, and billing information. HTML and CSS are used to customize the website's appearance and layout. Visual Studio Code is used for developing and managing the project files.

6. Example of Smart Bill Book Working:

For example, a customer wants to book an LPG cylinder through the Smart Bill Book website. The customer first logs in using their registered mobile number and password. After successful login, the customer can access the dashboard and submit a cylinder booking request. The system stores the booking details in the SQLite database. Agency employees can view and process the booking, update its status, and generate the required bill or invoice. This process reduces manual paperwork, saves time, and makes LPG booking more convenient.

9. Data Processing and Management:

Smart Bill Book uses Python and SQLite to process and manage gas agency information. When customers register, request LPG connections, book cylinders, or access billing services, the system processes the information and stores the relevant records in the database. Administrators and employees can retrieve and update records according to their assigned roles. The system supports customer management, booking processing, delivery tracking, automated billing, and report

generation. This centralized approach improves data organization, reduces manual errors, and supports efficient gas agency operations.

CHAPTER 4 ADVANTAGES AND DISADVANTAGES

4.1 Advantages:

1.Smart Bill Book reduces manual paperwork by digitalizing customer registration, LPG cylinder booking, billing, and gas agency management activities.

2.The system saves time by allowing customers to book LPG cylinders and access services online without visiting the gas agency.

3.Automated billing and invoice generation help reduce calculation errors and improve the accuracy of financial records.

4.Separate dashboards for administrators, employees, customers, and delivery personnel provide organized management and role-based access.

5.SQLite database integration allows customer information, booking details, payment records, and delivery information to be stored and retrieved efficiently.

6.The system improves transparency by allowing customers to check LPG connection requests, cylinder booking statuses, and billing information.

7.Smart Bill Book improves overall gas agency efficiency by simplifying daily operations, reducing repetitive tasks, and supporting better customer service.

4.2 Disadvantages:

1.Smart Bill Book requires a stable internet connection for users to access the web application when it is hosted online.

2.Users with limited computer knowledge may initially face difficulties understanding the website and its features.

3.Technical errors, server failures, or database problems may temporarily interrupt LPG booking and billing operations.

4.The system requires regular maintenance, database backups, and security updates to protect customer information and maintain reliable performance.

5.The current system depends on employees updating booking and delivery statuses correctly to maintain accurate records.

6.The SQLite database is suitable for small and medium-sized applications but may require upgrading to a more scalable database s

CHAPTER 5 CONCLUSION

Smart Bill Book is a web-based Gas Agency Management System developed using Python, Streamlit, and SQLite. The project aims to simplify and digitalize daily LPG gas agency operations, including customer registration, LPG connection management, cylinder booking, automated billing, and delivery tracking. The system provides separate dashboards for administrators, employees, customers, and delivery personnel. It helps reduce manual paperwork, improve data accuracy, save time, and maintain organized records of gas agency activities.

Smart Bill Book provides an efficient and user-friendly platform for managing LPG-related services through a centralized system. The project demonstrates the practical application of Python programming, database management, and web application development. In the future, additional Artificial Intelligence and Machine Learning features, such as LPG demand prediction, smart delivery planning, inventory forecasting, customer feedback analysis, and AI-based decision support, can be implemented to improve the system further. Overall, Smart Bill Book offers a foundation for modernizing gas agency operations, improving customer satisfaction, and supporting efficient business management.

CHAPTER 6 PROJECT INITIATION

6.1 Description of Problem Statement:

The traditional management of LPG gas agencies involves several manual processes, such as customer registration, cylinder booking, LPG connection management, billing, and delivery tracking. These activities require maintaining large amounts of paperwork and customer records, which can be time-consuming and difficult to manage. Manual record-keeping may lead to incorrect entries, duplicate bookings, billing errors, and delays in cylinder delivery. Customers often need to visit the gas agency or contact employees to book cylinders, check connection details, and obtain billing information. Therefore, a centralized digital system is required to simplify these operations, reduce manual work, and improve the overall efficiency of gas agency management.

Smart Bill Book is developed to overcome these problems by providing a web-based Gas Agency Management System using Python, Streamlit, and SQLite. The system allows customers to register, apply for LPG connections, book cylinders, upload documents, and access billing information through an online platform. Separate dashboards are provided for administrators, employees, customers, and delivery personnel to manage their respective activities. The system maintains customer records, booking details, employee information, payment records, and delivery updates in a centralized database. Additionally, future enhancements such as LPG demand prediction, smart delivery planning, and AI-based business analytics can further improve the efficiency and decisionmaking capabilities of the gas agency.

6.2 Scope of the Project:

The scope of Smart Bill Book is to develop a user-friendly and efficient web application for managing the daily operations of LPG gas agencies. The system covers customer registration, secure login, LPG connection requests, cylinder booking, document management, automated billing, payment records, and delivery management. Administrators can manage customer information, employee accounts, cylinder pricing, and agency reports, while employees can process bookings and update delivery records. Customers can access their dashboards to view LPG connection details, booking statuses, and available invoices. The project uses Python for application development, Streamlit for the web interface, and SQLite for database management.

The future scope of Smart Bill Book includes implementing Artificial Intelligence and Machine Learning techniques to improve gas agency operations. Proposed features include LPG demand

prediction using historical booking data, smart delivery scheduling, inventory forecasting, customer feedback and complaint management, and AI-based decision-support dashboards. These enhancements can help gas agencies predict cylinder requirements, identify delivery delays, monitor business performance, and make better operational decisions. The system can also be expanded to support multiple gas agency branches, online payment integration, SMS notifications, and advanced reporting facilities. These future developments aim to make Smart Bill Book a more intelligent, scalable, and reliable Gas Agency Management System.

Objectives of the Project:

The main objective of Smart Bill Book is to develop a web-based Gas Agency Management System that simplifies daily LPG agency operations. The project aims to reduce manual paperwork, improve customer service, maintain accurate records, and provide an efficient platform for managing LPG connections, cylinder bookings, billing, and delivery activities.

To provide an online customer registration and secure login system for accessing LPG gas agency services.

To simplify LPG connection applications by allowing customers to submit personal details and required documents online.

To provide separate dashboards for administrators, employees, customers, and delivery personnel according to their responsibilities.

To automate LPG billing and invoice generation while maintaining accurate payment and transaction records.

To manage customer information, employee records, cylinder bookings, and delivery details using an SQLite database.

To improve delivery management by allowing authorized employees to update cylinder booking and delivery statuses.

To provide administrators with access to sales reports, customer records, cylinder pricing, and daily agency operations.

To support future Artificial Intelligence and Machine Learning features for LPG demand prediction, inventory forecasting, and business analysis.

6.3 Stakeholders:

The stakeholders of Smart Bill Book are the individuals and organizations who use, manage, develop, or benefit from the Gas Agency Management System. They play important roles in ensuring the effective functioning and maintenance of the application.

1. **Customers:** Customers are the primary users of the system who register, apply for LPG connections, book cylinders, upload documents, and access billing information.

2. **Administrators:** Administrators manage gas agency operations, customer records, employee accounts, cylinder pricing, billing information, and business reports.

3. **Employees:** Employees are responsible for processing LPG connection requests, verifying customer documents, managing cylinder bookings, and updating service records.

4. **Delivery Personnel:** Delivery personnel are responsible for handling assigned LPG cylinder deliveries, updating delivery statuses, and maintaining delivery-related information.

5. **Gas Agency Owners and Managers:** Agency owners and managers use the system to monitor daily operations, customer activities, cylinder bookings, financial records, and overall business performance.

6. **Developers:** Developers are responsible for designing, developing, testing, maintaining, and improving the Smart Bill Book web application using Python, Streamlit, and SQLite.

7. **Project Guide and Evaluators:** Project guides and evaluators provide technical guidance, review project development, evaluate system functionality, and suggest improvements to enhance the project's overall quality.

6.4 Platform / Equipment / Resources:

1. Platform:

The Smart Bill Book application is developed using Python programming language and Streamlit framework. It can run on Windows or Linux operating systems through a web browser. SQLite is

used for storing and managing customer details, LPG cylinder bookings, billing information, payment records, and delivery details.

2. Equipment:

Computer or Laptop: Required for developing and running the application.

Processor: A standard modern processor suitable for Python development.

RAM: Minimum 4 GB recommended for running the application smoothly.

Internet Connection: Required for accessing online services and using the web application when hosted online.

Storage: Sufficient space for source code, database files, documents, and project resources.

3. Resources:

Resources	**Name**	**Purpose**
Operating System	Windows 11	To develop and run the application.
Web Framework	Streamlit	To create the web interface and dashboards.
Database	SQLite	To store customer, booking, billing, and delivery records.
Code Editor	Visual Studio Code	To write and debug program code.
Web Browser	Google Chrome	To access and test the application.
Package Manager	pip	To install required Python libraries.
Programming Language	Python	To implement application logic.

CHAPTER 7 SEMINAR PRESENTATION

S.S.V.P.S's B. S. Deore Polytechnic, Dhule Department of Computer Engineering *Academic Year : 2026-2027*		
SEMINAR AND PROJECT INITIATION COURSE		
Sem : CO 5K	**SPI – 315003**	**Date : 17/10/2026**
Topic : Smart Bill Book		
Roll No.	**Name of Student**	
88	Aarya Milind Shinde	
104	Mrudula Sachin Upakare	
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72	Samiksha Chatur Patil	

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1. Bibliography
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Abstract

The Smart Bill Book – Gas Agency Management System* is a web-based application developed using Python, Streamlit, and SQLite to simplify and automate the daily operations of LPG gas agencies. The system manages customer registration, LPG connection requests, cylinder bookings, billing, invoice generation, and delivery tracking.

It provides separate dashboards for administrators, employees, and customers to manage their activities efficiently. The application reduces paperwork, minimizes manual errors, improves data accuracy, and saves time. SQLite is used to store and manage customer details, booking records, billing information, and delivery status.

The main aim of this project is to provide a user-friendly, reliable, and efficient digital solution for gas agency management..

1. Introduction

- The Smart Bill Book is a web-based Gas Agency Management System designed to simplify LPG gas agency operations.
- It is developed using Python, Streamlit, and SQLite.
- The system manages customer registration and LPG connection requests.
- It provides facilities for cylinder booking and delivery tracking.
- The application automates billing and invoice generation.
- It maintains customer details, booking records, and payment information.
- The main aim is to reduce paperwork, minimize errors, and improve the efficiency of gas agency management.

2. Literature Survey

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	Computerized Gas Agency Management System	DBMS, Application Software	Customer, Stock, Billing Records	Computerized billing and stock management	Accurate data entry required
2	Automated Gas Agency Billing System	Database, Billing Software	Customers, Cylinder Prices, Bills	Simplify billing and reduce errors	Incorrect data causes billing errors
3	Gas Booking and Customer Management System	Web Application, Database	Customers, Bookings, Delivery Dates	Manage customers and bookings	Regular updates required
4	Online Gas Booking and Payment Integration	Web Technologies, Database, Payment Gateway	Bookings, Payments, Stock	Online booking and payment	Internet and payment security required
5	Inventory and Stock Management System	DBMS, Inventory Software	Cylinder Stock, Quantities	Track stock and availability	Incorrect stock entries
6	Customer Database Management System	Relational database, SQL	Customer IDs, names, addresses and contact information	To store customer information systematically and retrieve records quickly	Requires access control, data validation and database backups

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
7	Digital Invoice Generation System	Billing Software, Database	Invoice Numbers, Prices, Totals	Generate digital invoices	Incorrect pricing
8	Transaction Management System	Database, SQL	Payments, Dates, Bills	Maintain transaction history	Duplicate or incomplete records
9	Web-Based Gas Agency Management System	HTML, CSS, JavaScript, Database	Customers, Bookings, Stock, Billing	Centralized agency management	Security and maintenance required

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
10	Automated Bill Calculation System	Programming, Database, Calculation Logic	Price, Quantity, Charges	Automatic bill calculation	Incorrect input handling
11	Billing Report and Record Management System	Database, Reporting Tools	Daily Bills, Monthly Transactions	Generate billing reports	Accurate data required
12	Smart Bill Book for Gas Agency	Web Technologies, Database, Billing Logic	Customers, Bills, Payments, Bookings	Digital billing and record management	Testing and maintenance required

3. Information to seminar topic

- Technology: Python, Streamlit, and SQLite
- Main Purpose: To simplify and automate the daily operations of an LPG gas agency.
- Key Concept: Managing customer details, cylinder bookings, billing, and delivery records digitally.
- Approach: Provides a web-based system with separate dashboards for customers, administrators, and employees.
- Main Benefit: Reduces paperwork, minimizes manual errors, and saves time.
- Application: Can be used by LPG gas agencies to manage customers, bookings, invoices, and delivery status efficiently.

4. Advantages / Disadvantages

Advantages

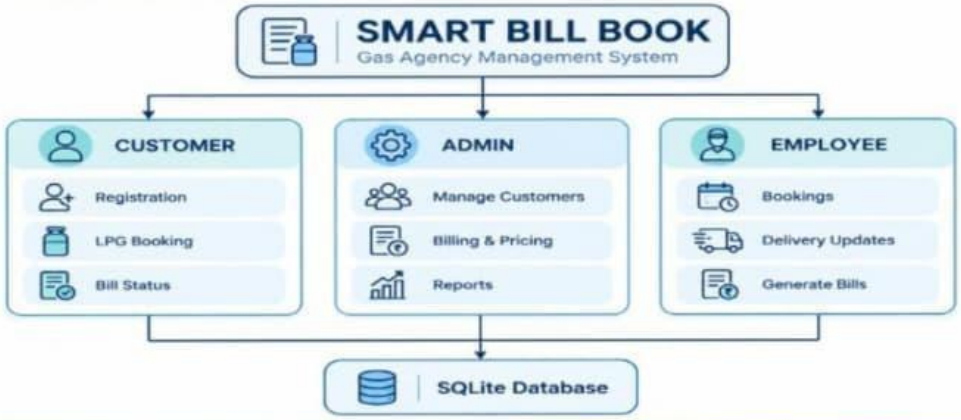
- Protects user's personal information.
- Reduces unnecessary sharing of sensitive data.
- Keeps important information needed for AI responses.
- Helps maintain the quality of AI-generated answers.
- Makes AI usage safer and more trustworthy.

Disadvantages

- May incorrectly identify personal information.
- Could hide details that are important for a response.
- Adds extra processing time before sending a prompt.
- Requires regular updates to improve detection accuracy.
- Cannot guarantee complete protection against every privacy risk.

5. Conclusion

The Smart Bill Book is a web-based Gas Agency Management System developed using Python, Streamlit, and SQLite. It simplifies customer registration, LPG cylinder booking, billing, and delivery management. The system provides separate dashboards for customers, administrators, and employees. It reduces manual paperwork, minimizes errors, and maintains accurate records. Overall, the system improves the efficiency, accuracy, and reliability of gas agency operations.



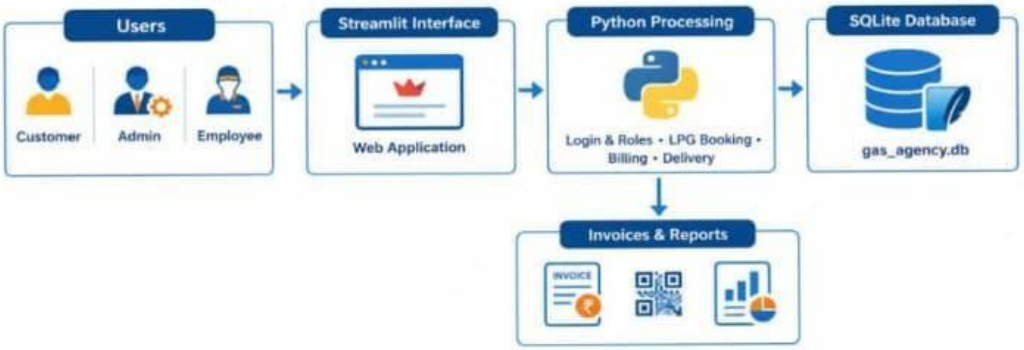
6. Project Initiation

The Smart Bill Book project was initiated to address the problems of manual billing, customer record management, and LPG cylinder booking in gas agencies. Traditional methods require paperwork and may cause calculation errors, delays, and difficulties in maintaining accurate records.

The project aims to develop a web-based Gas Agency Management System using Python, Streamlit, and SQLite. It automates customer registration, LPG booking, billing, invoice generation, and delivery tracking. The system provides separate dashboards for customers, administrators, and employees to manage their activities efficiently. As a result, it reduces paperwork, minimizes errors, saves time, and improves the overall efficiency of gas agency operations.

6. 1 Description of problem statement. identification.

The traditional gas agency system involves manual customer registration, LPG booking, billing, and delivery tracking, which can cause errors and delays. The Smart Bill Book system automates these activities through a web-based application. It uses Python, Streamlit, and SQLite to manage customer details, bookings, bills, and invoices. The system maintains organized records and reduces paperwork. It improves accuracy, saves time, and increases the efficiency of gas



6. 2 Scope and objectives.

Objectives

- Automate customer registration and LPG cylinder booking.
- Generate accurate bills and invoices.
- Manage customer details and booking records.
- Track cylinder delivery and payment status.
- Reduce manual work and calculation errors.
- Improve the efficiency of gas agency operations.

Scope

- Develop a web-based Gas Agency Management System.
- Use Python, Streamlit, and SQLite for development.
- Provide separate access for customers, administrators, and employees.
- Maintain customer, booking, billing, and payment records.
- Generate invoices and support delivery tracking.
- Simplify gas agency operations through centralized data management.

6. 3 Stake holder

- **End Users:** Individuals that will use the AI-powered tools and want to make sure
  their personal data is kept safe.
- **AI Service Providers:** The companies offering services of AI chatbots and large
  language models.
- **Developers:** The individuals that will design the AI-based Privacy Gateway.
- **Organizations:** The businesses that will be utilizing the AI tools and have concerns about the security of their data.
- **Privacy and Security Teams:** The personnel in charge of the well-being of the personal data within the organizations.
- **Researchers:** The individuals that will conduct research on the AI privacy tools and enhance their quality.

6. 4 Seminar Execution Plan

1. **Current Scenario:** Manual management of customer details, LPG bookings, billing, and deliveries.

2. **Privacy Challenges:** Billing errors, service delays, and difficulty maintaining records.

3. **Existing Approaches:** Paper registers and basic software for gas agency operations.

4. **Research Gap:** Lack of centralized data management and automated billing.

5. **Research:** Develop Smart Bill Book using Python, Streamlit, and SQLite.

6. **Future Impacts:** Reduce paperwork, improve accuracy, and provide faster service.

7. **Future Work:** Introduce online payments, SMS notifications, and real-time delivery tracking.

6.5 Platform/ Equipment/ Resources

Hardware Requirements:
- Laptop or desktop computer.
- Minimum 4 GB RAM (8 GB recommended).
- Stable internet connection.

Software Requirements:
- Operating System: Windows or Linux.
- Programming Language: Python.
- Framework: Streamlit.
- Database: SQLite.
- Code Editor: Visual Studio Code.

Other Resources:
- Customer details and LPG booking data.
- Billing and invoice generation modules.
- SQLite database for storing records.
- Testing tools to verify system functionality.

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- **Research Papers / Technical Resources**

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Thank you

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Database System Concepts — Abraham Silberschatz, Henry F. Korth and S. Sudarshan

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SEMINAR and PROJECT INITIATION Assessment / Evaluation

Sr. No	Enrollment Number	Seat Number	Name of student
1	24610960254	388559	Samiksha Chatur Patil
2	24610960274	388575	Aarya Milind Shinde
3	24610960286	388586	Vedshri Kamlesh Suryawanshi
4	24610960291	388591	Mrudula Sachin Upakare

Formative Assessment

Sr. No.	Criteria	Weightage	Exam Seat Number			
			388559	399575	388586	388591
1	SelectionTopic / Theme of seminar	5				
2	Literature review and data presentation	5				
3	Quality of Preparation and innovativeness	5				
4	Q-A handling	5				
5	Time Management	5				
6	Seminar Presentation report	10				
7	Selection of Theme of Problem Statement and its innovativeness	5				
8	Stages of development of Action plan	5				
9	Prototyping	5				
	Total	**50**				
	Scaled to 25					

Sign:

Name:Mr.Nishad.D.Patel

(Course Experts/s)